



DEBOPAM DUTTA

ASPIRING BIOTECH ENGINEER

CONTACT

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- KOLKATA , WB
- [portfolio](#)

SKILLS

- Python
- C
- ML & Deep Learning
- Computer Vision (yolo & opencv)
- IoT & Embedded systems
- Sensor Devices & RFID
- MATLAB, TensorFlow, PyTorch
- Time Management
- Effective Communication
- Adaptability & Fast Learning
- Problem-Solving
- PYMOL

LANGUAGES

- English (Fluent)
- Bengali (Fluent)
- Hindi (Intermediate)

INTERESTS

- IoT and Smart Devices
- AI in Healthcare & Biology
- Bioinformatics
- Computational Biotechnology
- RF Engineering & Wearable Technology

PROFILE

Aspiring Biotechnology Engineer (2nd Year, UEM Kolkata) passionate about combining biology with AI, IoT, and sensor technologies. Experienced in projects such as an AI-powered symptom-to-disease chatbot, a dual-factor attendance system with RFID + face recognition, and a bacteria counting system using computer vision. Skilled in Python, IoT hardware (Arduino, ESP 32, Raspberry Pi), and Hugging Face LLMs. A fast learner and persistent problem-solver, eager to apply emerging technologies to real-world challenges in healthcare and biology , eager to apply knowledge and gain practical experience.

EDUCATION

B.Tech in Biotechnology (2nd Year)

University of Engineering and Management

GPA: 8.67/ 10 (3RD SEM)

Higher Secondary (H.S.)

Subhaspalii Vidyaniketan (H.S.)

GPA: 8.5 / 10.0

PROJECTS

Wearable AI Navigation Assistant : - Finalist HST26 IIT Mandi

- Developed a real-time, offline navigation system using YOLOv8 for obstacle detection and Ollama for local scene interpretation.
- Improved detection accuracy by 67% by retraining models on a custom dataset tailored to complex Indian urban environments.
- Integrated Google Gemini API for interactive context and Python Text-to-Speech to provide actionable auditory feedback.

Dual-Factor Attendance System

- Built an IoT-based attendance tracker to prevent proxy attendance.
- Integrated RFID authentication with facial recognition for dual-factor security.
- The data can be sent directly to Google sheets for records.

Bacteria Colony Counting System

- Created a computer vision tool for automated bacterial colony counting. Designed to analyze lab images for biological research applications.

Automated Bacterial Identification & Antibiotic Resistant analyzer

- AI-powered system for automated bacterial identification and antibiotic resistance analysis using zone of inhibition data and machine learning.

EcoRemed AI : Bioremediation Site Analyzer

- An AI-powered environmental intelligence system that uses CNN and deep learning models to detect pollution, forecast contamination severity, and recommend optimized bioremediation strategies with explainable decision support. In progress.